MaxAir Release History

Current Version	**3.23**
Current Build	**091026**
Current Database Version	**1.0**
Current Database Build	**091026**

Last Update

- Commit #574 – Update – 'ajax_fetch_data.php', 'model.php. 'db.php, 'gateway.py' , 'request.js', '091026.sql and language files– add support to 'Sensor Types' model to allow setting of the number of decimal places displayed and to either add, subtract, multiply or divide the displayed value by a definable value.

- Commit #574 – New Feature – tuya_bridge.py' and 'check_tuya_bridge.php' – add script and schedule check support for Tuya WiFi relays.

- Commit #574 – New Feature – model.php' andlanguage files – add new 'Install Software' option for 'tinytuya' Python module.

- Commit #574 – Update – 'gpio_ds18b20.py' – improve error reporting and ensure and MySQL use of 'nodes.node_id' uses a text type parameter.

- Commit #574 – Update – 'rc10_control.py' and 'e7c_control.py' – for sensors where the value does not change, update the 'last_seen' once every 1 minute.

- Commit #574 – Update – 'ajax.php' – set pop-up graph title dependant on the sensor type.

History

Date	Ver	Build	DB Ver	DB Build	ChangeLog
28/10/2020					Ported from PiHome, initial HVAC build
30/10/2020					Improvements to setup.php
07/12/2020					Change controller.php to operate on state change
19/12/2020					Bug Fixes and performance improvements
20/12/2020					Performance improvements, change boiler table to system_controller
24/12/2020					Further updates, update demo databases
26/12/2020					Further updates, remove redundant files
26/12/2020					Udate system_c.php to work with Orange Pi boards
29/12/2020	0.1	Beta 1			Introduce code version and build numbering
03/01/2021	0.1	Beta 1			Further updates
04/01/2021	0.1	Beta1			Add Orange Pi Zero pin mapping
05/01/2021	0.1	Beta 1			Update chars to use sensors table
06/01/2021	0.1	Beta 1			Cosmetic changes
07/01/2021	0.1	Beta 1			Allow creation of 'Dummy' sensor nodes
07/01/2021	0.1	Beta 1			Allow graph placement or suppression
09/01/2021	0.1	Beta 1			Bug fixes – graph related
11/01/2021	0.1	Beta 1			Bug Fixes – graph shading
13/01/2021	0.1	Beta 1			Combine Zone and System Controller logs
14/01/2021	0.1	Beta 1			Add support for Orange Pi Zero Plus board
14/01/2021	0.1	Beta 1			Combine add-on log tables
15/01/2021	0.1	Beta 1			Bug Fix
16/01/2021	0.1	Beta 1			Bug Fix – Orange Pi board detection
20/01/2021	0.1	Beta 1			Graphing for Add_On controllers
20/01/2021	0.1	Beta 1			Change all references from 'boiler' to 'system_controller'
21/01/2021	0.1	Beta 2			Update to setup.php to add default network_settings record
22/01/21	0.1	Beta 2			Only process SonOff devices if network is available
29/01/2021	0.1	Beta 2.1			Bug Fix and update to Onetouch menu
03/02/2021	0.1	Beta 2.1			Update to livetemp
04/02/2021	0.1	Beta 2.1			Improvements to livetemp
06/02/2021	0.1	Beta 2.1			Bug Fix – onetouch
08/02/2021	0.1	Beta 2.2			Relay control managed using the messages_out queue, improvements to livetemp

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10/02/2021	0.1	Beta 3			Schedule control using Python Schedule

					in place of Cron
11/02/2021	0.1	Beta 3.2			Implement GUI for Job Scheduling
12/02/2021	0.1	Beta 3.2			Bug Fix – controller.php
15/02/2021	0.1	Beta 4			Change to Frost Protection to make it sensor rather than zone based
16/02/2021	0.1	Beta 4			Use node_id in relay setup and display
17/02/2021	0.1	Beta 4			Bug Fix – email address entry
17/02/2021	0.1	Beta 4			Bug Fix – email address modal
17/02/2021	0.1	Beta 4			Bug Fix – Frost Protection
18/02/2021	0.1	Beta 4.2			User Account Management
18/02/2021	0.1	Beta 4.2			Enable/Disable User Accounts
18/02/2021	0.1	Beta 4.2			Enable the disabling of a looged-on user account
19/02/2021	0.1	Beta 4.3			Add admin category to User Accont Management, suppress Settings menu for non-admin users. Delete redundant files
19/02/2021	0.1	Beta 4.3			Tidy up display of disabled delete buttons
20/02/2021	0.1	Beta 4.3			Bug Fix – udate_db.php referencing the wrong repository
20/02/2021	0.1	Beta 4.3			Bug Fix – boostSet.php
20/02/2021	0.1	Beta 4.3			Tidy up Views
20/02/2021	0.1	Beta 4.3			Bug Fix – setup.php
20/02/2021	0.1	Beta 4.3			Re-commit of model.php
21/02/2021	0.1	Beta 4.3			Improvements to Job Scheduling
21/02/2021	0.1	Beta 4.3			Implement System Controller Max Run Time
21/02/2021	0.1	Beta 4.4			Big Fix – Add relay modal, reorder Settings Menu
22/02/2021	0.1	Beta 4.4			Tidy up of model.php
23/02/2021	0.1	Beta 4.4			Re-sync messages_out on gateway startup, HVAC bug fix for frost protection and update to test databases
23/02/2021	0.1	Beta 4.4			Update relay.php to include select/option applied to sensor.php
24/02/2021	0.1	Beta 4.4			Bug Fix – header.php
25/02/2021	0.1	Beta 4.4			Bug Fix – add/edit System Controller record
25/02/2021	0.1	Beta 4.4			Add Status Menu option to display last Job Schedule report
26/02/2021	0.1	Beta 4.5			Add option to gateway to not process networked messages_out (test mode)
27/02/21	0.1	Beta 4.5			Add python script to update code base from GitHub repository

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27/02/2021	0.1	Beta 5			Rename database from ‘pihome’ to ‘maxair’
27/02/2021	0.1	Beta 5			Home Assistant Integration – api changes
05/03/2021	0.1	Beta 5			Bug Fix – add additional zone controllers
06/03/2021	0.1	Beta 5			Enable blocking of outgoing messages (test mode)
07/03/2021	0.1	Beta 5			Bug Fix – onetouch.php
14/03/2021	0.1	Beta 5			Bug Fix – Raspbian compatibility for AutoHotSpot
17/03/2021	0.1	Beta 5			Bug Fix – work in AP mode
17/03/2021	0.1	Beta 5			Bug Fix – Echo and HomeKit installer
18/03/2021	0.1	Beta 5			Improvements in Echo and HomeKit installer
20/03/2021	0.1	Beta 5			Bug Fix – allow scheduling of zones with no attached sensor (Tasmota)
22/03/2021	0.1	Beta 5			Bug Fix – gateway.py, enable/block outgoing messages. Bug Fix – add new scheduled job
23/03/2021	0.1	Beta 5			Implement WiFi reboot for systems running NetworkManager
24/03/2021	1.0	RC1			Move to Release Candidate 1
26/03/2021	1.0	RC1			Move AutoHotSpot to add_on directory
26/03/2021	1.0	RC1			Add nodes listing to Status menu
26/03/2021	1.0	RC1			Update to English language file
27/03/2021	1.0	RC1			Add documentation
27/03/2021	1.0	RC1			Add documentation Tasmota Lamps
27/03/2021	1.0	RC1			More documentation
28/03/2021	1.0	RC1			Update to Operating Modes documentation
28/03/2021	1.0	RC1			Documentation – add separate Boiler and HVAC mode setup guides
29/03/2021	1.0	RC1			Documentation – Live Temperature Control, Displaying Graphs and Frost protection
30/03/2021	1.0	RC1			Update to APIs and API Documentation
31/03/2021	1.0	RC1			Bug Fix – create HTTP message and Documentation
31/03/2021	1.0	RC1			Zone Types Documentation
31/03/2021	1.0	RC1			Task Scheduling Documentation
01/04/2021	1.0	RC1			Add AutoHotSpot to Services Configuration menu
02/04/2021	1.0	RC1			Improve install of APIs – add script to check for and enable Apache2 mod-rewrite

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08/04/2021	1.0	RC1			Enable background add-on install, plus

					Bug Fixes
08/04/2021	1.0	RC1			Big Fix – sw_install.py. Display install text when finished installing
10/04/2021	1.0	RC1			Bug Fix – setup.php. New version of software install process.
11/04/2021	1.0	RC1			Documentation – Software Install
11/04/2021	1.0	RC1			Documentation – Technical description of install process. Add documentation access from the System Status menu
12/04/2021	1.0	RC1			Documentation Update – Task Scheduling. Add gpio_ds18b20 disabled task, all other tasks enabled by default by setup.php
13/04/2021	1.0	RC1			Documentation – Using Dummy Sensors to Import External Data
13/04/2021	1.0	RC1			Bug Fix – Gateway type default set to Serial if gateway table is empty
14/04/2021	1.0	RC1			Add Si7021 Humidity Sensor, with selection of sensor type in sensor.php, plus associated database changes
16/04/2021	1.0	RC1			Improvement to task scheduling to allow once daily jobs. Enable db_cleanup task. Simplification of humidity graphing. Bug Fix
16/04/2021	1.0	RC1			Update to controller.php for OFF mode to set system completely OFF (NO frost protection)
17/04/2021	1.0	RC1			Documentation – Humidity Sensors
27/04/2021	1.0	RC1			Change table temperature_sensors to sensors and controller_relays to relays. Database changes can be implemented by executing the php script ‘migrate_sensors_relays.php from within the /var/www/MySQL_Database directory
28/04/2021	1.0	RC1			Add Sensor Type GUI. Terminate gateway script if route to network gateway is lost
01/05/2021	1.0	RC1			Enable deletion of unused nodes
02/05/2021	1.0	RC1			Bug Fix – db.php
02/05/21	1.0	RC2			Remove Nodes optionfrom System Status menu as function now provides by Nodes in Node and Zone Configuration menu. Add tooltips to Sensors and Relays model for disabled entries

Date	**Ver**	**Build**	**DB Ver**	**DB Build**	**ChangeLog**
04/05/2021	1.0	040521			First Release Version Add hardware support for the PineH64 board. Manage messages_out queue entries from relay.php rather than zone.php

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06/05/2021	1.0	050521			Enable Zone delete at the end of controller.php cycle using purge record marker (this function was previously performed by piconnect.php)
06/05/2021	1.0	050521			Tidy up Zone Deletion. Documentation
07/05/2021	1.0	050521			Bug Fix – update mysql_database.sql Foreign Key Constraint name on relay table
08/05/2021	1.0	080521			Database update for zone_controllers and relays tables. Associated php modules code updates. Note: DO NOT run update_db.php, instead use migrate_zone_controllers.php to retain existing data
11/05/2021	1.0	080521			Bug Fix – database
11/05/2021	1.0	080521			Add check to migrate_zone_controllers.php to abort if zone_relays table already exists
18/05/2021	1.0	170521			Bug Fix – Boost Update deleting GIO controller messages_out records when no boost console is installed
22/05/2021	1.0	210521			Bug Fix – Correct Login heading when AutoHotSpot is not installed. DS18B20 Documentation
25/05/2021	1.0	250521			Add support for Rock Pi E board
08/06/2021	1.0	250521			Update Pin_Dict.py add Orange Pi Zero2 1x13 header GPIO pins
08/06/2021	1.0	250521			Tidy up – removing duplicates
09/06/2021	1.0	250521			LoRa (RFM95) sketch
09/06/2021	1.0	250521			Add documentation for using the Orange PI Zero 2 board
10/06/2021	1.0	100621			Update db_cleanup.php. Remove delete zone_graphs as already actioned in gateway.py. Update system_c.pjp board detection method to use Adafruit PlatformDetect. Correct latest document name spelling.
10/06/2021	1.0	100621			Documentation – Adafruit Blinka

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12/06/2021	1.0	100621			Update Orange Pi Zero 2 documentation – implementing 1-wire interface
13/06/2021	1.0	100621			Delete document, in wrong directory
16/06/2021	1.0	100621			Add separate AutoHotSpot function for distributions running NetworkManager
17/06/2021	1.0	170621			Bug Fix = installer improvements and implementation of iptables for connection with internet
18/06/2021	1.0	170621			Update README.md to include list of supported boards
28/06/2021	1.0	190621			Update build number
28/06/2021	1.0	190621			Commenting out update title
27/07/2021	1.0	270721			Trap error line in gateway.py Tidy up Pin_Dict.py
29/07/2021	1.0	280721			Gateway.py – if WiFi type gateway then ping router once every 3600 seconds
30/07/2021	1.0	300721			Bug Fix – code update
04/08/2021	1.0	010821			Move MaxAir Update to the Settings.Status meni and rename as MaxAir Versions
05/08/2021	1.0	040821			Add Update Code from GUI
06/08/2021	1.0	060821			Bug Fix – update_code.py for nested sub-directories
09/08/2021	1.0	070821	1.0		Bug Fix – move_files.php Initial implementation for database updates from GUI
11/08/2021	1.0	100821	1.0	BASE	Tidy up
12/08/2021	1.0	120821	1.0	BASE	Add documentation links to modal headers
14/08/2021	1.0	130821	1.0	130821	Implement Correction Factor for sensor readings Bug Fix – gpio_ds18b20.py
14/08/2021	1.0	140821	1.0	130821	Bug Fix – apply_database_update.php and setup.php to add any database updates required over the BASE database image
14/08/2021	1.0	150821	1.0	130821	Bug Fix – header.php not suppressing updates.txt in file list GPIO relay test
16/08/2021	1.0	170821	1.0	130821	Enable 1-wire script to return either Centigrade or Fahrenheit depending on units selected
16/08/2021	1.0	170821	1.0	170821	Bug Fix – add column sub_type to nodes table to support humidity sensors
17/08/2021	1.0	170821	1.0	170821	Fixes for working in Fahrenheit
18/08/2021	1.0	180821	1.0	180821	Allow selection of GPIO Relay trigger level

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19/08/2021	1.0	190821	1.0	180821	Update to move_files.php to kill gateway.py and/or gpio_ds18b20.py and/or jobs_schedule.py if they have been updated. Set live temp icon depending on Units. Documentation
23/08/2021	1.0	230821	1.0	180821	Bug Fix – scheduling.php, slider max temperature Fahrenheit conversion if required
26/08/21	1.0	240821	1.0	180821	Only allow 'admin' user to perform updates Sensor and Zone documentation
27/08/2021	1.0	260821	1.0	180821	Update foreign language files
28/08/2021	1.0	270821	1.0	270821	Allow creation of non temperature zones
28/08/2021	1.0	270821	1.0	280821	Update to 270821.sql to check zone type record does not already exist
30/08/2021	1.0	300821	1.0	300821	Enable definition of units character (or special character string) for Sensor Type
30/08/2021	1.0	300821	1.0	300821	Update Sensors documentation. Check for existing column in 300821.sql Bug Fix and improvements to setup.php
31/08/2021	1.0	310821	1.0	300821	Bug Fix for initial setup when no relays are installed find_my_gateway not triggering when incorrect ip is entered for wifi gateway
31/08/2021	1.0	310821	1.0	300821	Update check_gw.php to improve reliability of counter Ug Fix – zone.php when only 1 sensor is available
02/09/2021	1.0	010921	1.0	310821	Fix for add new node Fix for gateway.py to enable it to run when relays have been added but not allocated to a zone Basic Fault Finding documentation Improved document listing
02/09/2021	1.0	020921	1.0	310821	Documentation update. Missing model.php code update Change to update_code.py to fix directory privileges error introduced by using 'shutil'
03/09/2021	1.0	030921	1.0	310821	Bug Fix – remove shutil from update_code.py completely, use rm -R to avoid permission error
04/09/2021	1.0	040921	1.0	310821	Add sketch for Switch Sensor Sensor Types documentation Fix move_files.php

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06/09/2021	1.1	040921	1.0	310821	Bug Fix – change move_files.php to move_files.py
06/09/2021	1.1	060921	1.0	310821	Fix typo in documentation
06/09/21	1.1	060921	1.0	310821	Add 'Switch' zone, for use with a binary

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					sensor Documentation Change Arduino Sketches directory Bug Fix – update_code.py
06/09/2021	1.1	060921	1.0	310821	MySensors Gateway installation script
07/09/2021	1.1	070921	1.0	310821	Change to move_files.py to create any missing/new sub-directories
07/09/2021	1.1	070921	1.0	310821	Change default password to maxair
08/09/2021	1.1	070921	1.0	310821	Enable log pipe for MySensors Gateway
09/09/2021	1.1	070921	1.0	310821	Script to enable 1-wire sensors
11/09/2021	1.1	110921	1.0	310821	Switch Zone – Bug Fix Move switch zone to Add-On area of home screen
13/09/2021	1.1	130921	1.0	310821	Bug Fix – Switch Zone
13/09/2021	1.1	130921	1.0	310821	HA Integration
13/09/2021	1.1	130921	1.0	310821	Bug Fix – scheduling.php updated to match last table update
14/09/2021	1.1	140921	1.0	310821	Update ReadMe
14/09/2021	1.1	140921	1.0	110921	Move all add-on zones to right of boiler tile Change Lamp icon to Switch icon Separate out switching type zone processing in controller.php
16/09/2021	1.1	160921	1.0	110921	Modify Software Install Process to display name and description
16/09/2021	1.1	160921	1.0	110921	Stand-alone sensors
17/09/2021	1.1	170921	1.0	110921	Bug Fix – zone index_id not updating Add name/description to HomeAssistant install.sh
18/09/2021	1.1	180921	1.0	180921	Add Sunset/Sunrise option to scheduling.php Active Schedule Processing move to a function and processed using variable comparison rather than SQL time functions. Fixes a midnight rollover, where end time is on the day after start time. Bug Fix – not creating database backup
19/09/2021	1.2	180921	1.0	180921	Bug Fix – Update 180921.sql
19/09/2021	1.2	180921	1.0	180921	Bug Fix – Update relay.php, widen scope of MySensor name check
20/09/2021	1.2	200921	1.0	180921	Fix for model.php to add default value to gateway timeout, to prevent setup record not being save to database
21/09/2021	1.2	200921	1.0	180921	MQTT Nodes
22/09/2021	1.2	210921	1.0	180921	Partial bug fix controller.php to follow
22/09/2021	1.2	210921	1.0	180921	Bug Fix controller.php for MySensor relays
23/09/2021	1.2	210921	1.0	180921	Restore gateway.py
23/09/2021	1.2	210921	1.0	180921	Minor bug fix to controller.php for screen message using wrong variable name

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23/09/2021	1.2	210921	1.0	180921	Bug Fix
24/09/2021	1.2	210921	1.0	180921	Stabilization Fix
24/09/2021	1.2	240921	1.0	180921	Bug Fix for scheduling.php to display status icons correctly controller.php add missing variable initialization
26/09/2021	1.2	250921	1.0	250921	Update gateway.py for MQTT Nodes
30/09/2021	1.2	250921	1.0	250921	Use of column description instead of fix index access the cursor
03/10/2021	1.2	250921	1.0	031021	Bug Fix – Update gateway.py to reduce wifi ping check to once ever 60 seconds else network connection could be lost and restored with no gateway reset
07/10/2021	1.2	280921	1.0	031021	MQTT Web Interface
08/10/2021	1.2	081021	1.0	031021	Mode Button can operate in current cyclic mode or with pop-up button menu. Choose at Mode Selection either cyclic or button
09/10/2021	1.2	091021	1.0	031021	Bug Fix – mode.php
09/10/2021	1.2	091021	1.0	031021	Bug Fix – model.php to set correct value for Mode selection Fix for homelist.php where multiple zone controllers are attached to a single zone
10/10/2021	1.2	101021	1.0	031021	Home Assistant Integration Update
10/10/2021	1.2	101021	1.0	031021	Change to update_code.py in order to retain correct file permissions Add ChangeLog History document
10/10/2021	1.2	101021	1.0	031021	MQTT Nodes bug fix
12/10/2021	1.2	121021	1.0	031021	Bug Fix – move_files.py to fix version and build update to system table. Add pkill to maxair sudoers.d file (Note : added to setup .php but will require manual update to /etc/sudoers.d/maxair to add ‘www-data ALL=(ALL) NOPASSWD:/bin/pkill’
13/10/2021	1.3	131021	1.0	031021	Add Support for MQTT Devices

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15/10/2021	1.3	151021	1.0	031021	Bug Fix to display Fahrenheit temperatures in Boost and Override Bug Fix display Boost and Override icons for both cyclic and button modes
17/10/2021	1.3	171021	1.0	031021	Bug Fix to model.php mode settings vale corrected for Button Control modes.
18/10/2021	1.3	181021	1.0	031021	Update to scheduling to not display Sunrise/Sunset options if Openwaether has not updated in the last 24 hours. Addition of Slovakian language file (thanks to Mr.T) Added support for MQTT sensors sending data in a nested JSON format
19/10/2021	1.3	191021	1.0	191021	Enable placement of selected Onetouch buttons on either Home or Onetouch screen
20/10/2021	1.3	201021	1.0	191021	Working in Fahrenheit Bug Fix for deadband temperature and to display cut-in and cut-out temperatures in Fahrenheit.
21/10/2021	1.3	211021	1.0	191021	Add GUI for Button page selection. Move db-cleanup.php interval values to new table db_cleanup.
22/10/2021	1.3	221021	1.0	191021	Bug Fix de.php error at line 176 Update to ajax.php and model.php to use language file for all labels. Update to Slovakian language file.
25/10/2021	1.3	251021	1.0	191021	Check for messages_in record. Split System Configuration menu and add new System Maintenance menu. Add GUI for db_cleanup table Add new table ‘graphs’ to allow selection of graph menus. Add GUI for modification of table ‘graphs’. Update ajax.php to use language files
26/10/2021	1.3	261021	1.0	261021	Update to notice.py to enable SMTP Port to be set, for SSL or non-encrypted operation. Update to E-Mail GUI.
27/10/2021	1.3	271021	1.0	261021	Fix for notice.py when interval is set to 0. Fix for start_backup.py to use correct FROM, TO addresses.
28/10/2021	1.3	281021	1.0	281021	Add battery level and threshold values to display output. Disable PiConnect option in Settings. Tidy-up system table for updates, use single field ‘repository’ to point to the GitHub repository. Add GUI to enable change. Change update_code.py to use repository value from system Table.
28/10/2021	1.3	281021	1.0	281021	Bug Fix model.php, not updating Alert

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					settings. Missed update_code.py in #257 Update to request.js when using document.getElementsByTagName("input") or that it operates within a DIV rather than over the whole of model.php
30/10/2021	1.3	301021	1.0	301021	Improvement to Repository selection.
31/10/2021	1.3	311021	1.0	311021	Tidy-up Repository selection, everything now contained in a single table 'repository' rather than using 'fork' and 'system' table. Bug Fix 200921.sql. Update to setup.php to enable re-install when database already exists.
01/11/2021	1.3	011121	1.0	311021	Further improvement to setup.php for when installing where no previous database copy exists.
04/11/2021	1.3	041121	1.0	311021	Use 20% wider buttons when de or sk language files are in use. Allow creation of dummy controller device (for systems which do not use a zone controller). Bug Fix for Reboot and Shutdown – use Python3 rather than Python2.
06/11/2021	1.3	061121	1.0	061121	Update to allow CPU Temperature at which notice is sent.
07/11/2021	1.3	071121	1.0	061121	Update to allow operation of gateway.py when no physical gateway exists (eg for systems with GPIO relays and 1-wire sensors).
08/11/2021	1.3	081121	1.0	061121	Bug Fix for homelist.php, not showing correct temperature for add=on typ zones.
09/11/2021	1.3	091121	1.0	061121	Update to sk language file Bug Fix to System Controller model.php to display currently selected relay correctly in the dropdown lists for heat, cool and fan relays. Documentation – Gateway Types and Setup

Date	Ver	Build	DB Ver	DB Build	ChangeLog
11/11/2021	1.3	111121	1.0	061121	Update ajax.php, modify Services menu to remove pihome_mqtt.service and replace with HA_integration.service.
12/11/2021	1.3	121121	1.0	061121	Update to Slovakian language file, thank you Mr.T. Bug fix gateway.py not updating last_seen for MQTT sensors.
13/11/2021	1.3	131121	1.0	061121	Bug Fix ajax.php not selecting set Type value when in Edit MQTT Connection.
15/11/2021	1.3	151121	1.0	061121	Add documentation for MQTT Connection Setup and Configuration.
16/11/2021	1.3	161121	1.0	161121	Change GPIO Output type gateway to Virtual as it better reflects its use.
18/11/2021	1.3	181121	1.0	161121	For SWITCH type zone, update home screen status on button press.
21/11/2021	1.3	211121	1.0	211121	Force gateway script to exit if MQTT disconnects. setup.php will create a 'Virtual' gateway as default. Bug Fixes for HVAC Mode operation. Update to Boiler and HVAC Setup Guides.
24/11/2021	1.3	241121	1.0	211121	Various documentation updates. controller.php additional error trapping (no functional change).
25/11/2021	1.3	251121	1.0	211121	Enable graphing of sensor data when in HVAC mode.
27/11/2021	1.4	271121	1.0	261121	Introduce new zone category HVAC-M to support multi-zone HVAC systems. Introduce 3 timer modes for HVAC systems, HEAT, COOL and AUTO. Add hvac_relays_state to table system_controller to enable home screen status updates of system controller button. Change 'look' of Modes button menu to fit extra line of text. Bug Fix schedulelist.php not showing not included zones correctly.
02/12/2021	1.4	021221	1.0	261121	Update sb-admin-2.css to add space above button.
04/12/2021	1.4	041221	1.0	261121	Enable swapping of heating and Hot Water graph positions.
08/12/2021	1.4	081221	1.0	261121	Bug Fix HVAC Mode, cool and fan relays were not being added to the messages_out queue.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
13/12/2021	1.4	131221	1.0	261121	Do not process zone overrun until a controller_zone_logs record exists.
15/12/2021	1.4	151221	1.0	261121	Bug Fix – Time Zone change will force a restart of gateway.py and if required gpio_ds18b20.py model.php correct spelling for Timezone Europe/Kiev Documentation update MQTT Setup Guide.
17/12/2021	1.4	171221	1.0	261121	Remove remaining instances of MySQL Now() function.
19/12/2021	1.4	191221	1.0	261121	Bug Fix - controller.php, count controller_zone_logs record count.
21/12/2021	1.4	211221	1.0	261121	Encrypted SMTP password stored in database. Encrypt MQTT connection password. Added password decryption to gateway.py(MQTT password), start_backup.py and notice.py, Added notice.py (disabled) to jobs table.
22/12/2021	1.4	221221	1.0	221221	Bug Fix for Override function, plus documentation.
23/12/2021	1.4	231221	1.0	221221	Bug Fix for start_backup.py and notice.py, missing newline before HOST =. Bug Fix for zone.php when more than 2 relays are attached.
24/12/2021	1.4	241221	1.0	221221	Bug Fix Email and MQTT re-encrypting already encrypted passwords, when record re-saved.
01/01/2022	1.4	010122	1.0	010122	New Feature to enable an start time offset to be applied to any selected schedule.
02/01/2022	1.4	020122	1.0	010122	Add Start Time Offset value to Offset button, when active. Fix button title font for One Touch buttons.
03/01/2022	1.4	030122	1.0	010122	Limit Start Time Offset to Maximum value when control temperature is less than low limit
05/01/2022	1.4	050122	1.0	010122	Add GPIO pin mapping for Orange Pi 3 and Orange Pi 3 LTS. Add Fahrenheit temperature span to Add Offset modal.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
09/01/2022	1.4	090122	1.0	010122	Bug Fix for HA_HVAC_integration.py, decrypt MQTT_PASSWORD. Add documentation for Zigbee2mqtt device connection.
10/01/2022	1.4	100122	1.0	010122	Supress Sunset/Sunrise option for Night Climate scheduling. Add documentation for Night Climate Scheduling.
12/01/2022	1.4	120122	1.0	010122	Add documentation for Orange Pi 3 LTS. Change which documentation dropdowns show Night Climate Scheduling.
14/01/2022	1.4	140122	1.0	010122	Bug Fix - HA_HVAC_integration.py, set system controller mode.
15/01/2022	1.5	150122	1.0	010122	Add new feature ‘Pump Relays’ for control of external pumps. Update zone_view to show relays.type field.
17/01/2022	1.5	170122	1.0	010122	Bug Fix for db.php, missing field when adding new controller.
21/01/2022	1.6	210122	1.0	210122	New Feature – add Email notification is sensor values are above or below pre-set values. Bug Fix for zone.php, issue with sp_deadband when converting to Fahrenheit.
23/01/2022	1.6	230122	1.0	230122	Remove Weather Factoring from HVAC mode. Make Weather Factoring optional for Boiler Mode.
24/01/2022	1.6	240122	1.0	240122	Allow selection of sensor to be used for Weather Factoring.
25/01/2022	1.6	240122	1.0	240122	Add missing notice.py for commit #304.
26/01/2022	1.6	260122	1.0	240122	gateway.py - apply Relay Trigger setting to MQTT type relays.
27/01/2022	1.6	270122	1.0	240122	Bug Fix – scheduling.php, fix for night climate update to operate on the correct record id.
28/01/2022	1.6	280122	1.0	240122	Bug Fix – Night Climate tile Status indicator not showing correct colour.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
29/01/2022	1.6	290122	1.0	240122	Maximum Operating Time and Hysteresis Time removed from Zone GUI. Note disabled in the GUI, database fields will be populated with default values, in case it is decided to reimplement at some future date.
30/01/2022	1.6	300122	1.0	240122	Bug Fix – getTemperature.php not using

					the correct node_id when accessing messages_in table. Move legacy database utility files into a new legacy folder within MySQL_Database folder. Bug Fix – gateway.py, only try and retrieve trigger levels for relay nodes.
31/01/2022	1.6	310122	1.0	240122	Bug Fix –schedulelist.php, sunset/sunset offsets not being retrieved from the database.
01/02/2022	1.6	010222	1.0	240122	Bug Fix – sw_install.py and HomeAssistant/install.sh
02/02/2022	1.6	020222	1.0	240122	Add documentation for Home Assistant Integration.
03/02/2022	1.6	030222	1.0	240122	Bug Fix for sensor.php, not updating correctly.
04/02/2022	1.6	040222	1.0	240122	Update to gateway.py, detect_chip_board.py and system_c.php to enable them to work with boards that do not support Adafruit Blinka for GPIO functionality. Note:- GPIO is disabled for these boards.
06/02/2022	1.6	060222	1.0	240122	Bug Fix – zone.php - enable editing of Boost Console Id Bug Fix – zone.php – only allow selection of unallocated sensors. Bug Fix – zone.php – fix displaying of unwanted fields Bug Fix – homekit/config_jason.py – reflect new database structure.
07/02/2022	1.6	070222	1.0	240122	Update to the Homekit add-on to remove the update function from within controller.php and replace it with a python script, which will be installed as a service as part of the overall Homebridge installation.
08/02/2022	1.6	080222	1.0	240122	Bug Fix – zone.php, missing sensor when in edit mode.
09/02/2022	1.6	090222	1.0	240122	Update to HomeKit add-on and Home Assistant Integration documentation.
09/02/2022	1.6	090222_2	1.0	240122	Update to HomeKit add-on. New api, thermostatscript.php to support Homebridge-http-webhooks thermostat. Home Assistant Integration documentation.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
10/02/2022	1.6	100222	1.0	240122	Update to webhooks.py to handle outlets type and to reflect MaxAir Boost and Switch Zone status back to Homebridge screen.
11/02/2022	1.6	110222	1.0	240122	Bug Fix – notice.py, not clearing messages when notice_interval = 0
11/02/2022	1.6	110222	1.0	110222	Away Scheduling, create a single schedule which will be executed when 'Away' is enabled.
11/02/2022	1.6	110222+2	1.0	110222_2	Bug Fix – controller.php to fix issues with logs, introduced in #325 Bug Fix – 110222.sql re-order queries
12/02/2022	1.6	120222	1.0	110222_2	Update to Away function so that it will operate irrespective of current Mode setting. Change to controller.php to use colour indication is schedule name is running or idle. Add documentation to describe Away Function. Add document to describe Zone Mode numbering.
13/02/2022	1.6	130222	1.0	110222_2	Disable Away function if in OFF mode. Use Away icon in zone tile if an Away schedule is running.
15/02/2022	1.6	150222	1.0	110222_2	Bug Fix – scheduling.php, coop option not displayed when working in menu type mode selection. Update HA Integration documentation.
16/02/2022	1.6	160222	1.0	110222_2	Add coop indicator to hone screen zone modal. Update to controller.php for switch zones, enable override while a schedule is running. Add zone name to controller.php display and re-arrange fields in zone displays.
16/02/2022	1.6	160222	1.0	110222_2	Updates to Switch type zone control using Amazon Echo and Home screen button to toggle state.
17/02/2022	1.6	170222	1.0	110222_2	Further updates to Switch type zone control using Amazon Echo and Home screen button to toggle state.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
18/02/2022	1.6	180222	1.0	110222_2	Performance improvement to Home screen load time by replacing schedule_daily_time_zone_view with a query.
20/02/2022	1.6	200222	1.0	110222_2	Bug Fix 1 db.php missing field in database update for system_controller table. Change to controller.php to allow changes to HTTP Tasmota switch state to be reflected in GUI.
22/02/2022	1.6	220222	1.0	110222_2	Performance updates to check schedule routine. Removal of screen print functions from controller.php to improve performance. Updates to Away Scheduling.
23/02/2022	1.6	220222	1.0	230222	Update to schedule_daily_time_zone_view to improve performance.
24/02/2022	1.6	240222	1.0	240222	Bug Fix - schedulelist.php to display sunrise/sunset icon in schedule list, if enabled. Performance improvement opening settingslist.php.
25/02/2022	1.6	250222	1.0	240222	Bug Fix – scheduling.php – not getting correct time zone status when editing. Fix for gateway.py when HTTP relay cannot be reached by IP address
28/02/2022	1.6	280222	1.0	240222	Bug Fix – Home screen displaying the wrong Away status icon when Away enabled, with no associated Away schedule.
03/03/2022	1.6	030322	1.0	240222	Update to RFM96 Sketches to fix issue with IRQ pin definition when using MySensors ver 2.3.2 library.
06/03/2022	1.6	060322	1.0	240222	Update to the Sensors option in the System Status menu. The last date time value is now a button, which when clicked on will display the timestamps for all sensors attached to the node over the last 24 hours.
07/03/2022	1.6	070322	1.0	240222	Update to the Sensors option in the System Status menu to add average count per sensor per hour.
08/03/2022	1.6	080322	1.0	080322	Update to Boost modal to allow deletion of zone boost setting when there is more than on setting present.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
09/03/2022	1.6	090322	1.0	080322	Bug Fix – controller on times, for home screen and usage graphs.
10/03/2022	1.6	100322	1.0	080322	Add gateway sketch for the wireless-tag WT32_ETH01 combined ESP32 WiFi and Ethernet board. Sketch is configurable for NRF24L01, RFM69 and RFM95 radio modules. Implements WiFiManager and supports Ethernet only working mode.
10/03/2022	1.6	100322_2	1.0	080322	Modification to gateway sketch for the wireless-tag WT32_ETH01 to add option to disable the ethernet interface using a jumper.
11/03/2022	1.6	110322	1.0	080322	Simplification of the gateway sketch for the wireless-tag WT32_ETH01
12/03/2022	1.6	120322	1.0	080322	Simplification of the gateway sketch for the wireless-tag WT32_ETH01 by redefining SPi pins within the sketch and starting the SPi interface before the MySensors radio.
18/03/2022	1.61	180322	1.0	080322	Bug Fix - MaxAir_GatewayW5100_v100.ino change MY_RADIO_NRF24 to MY_RADIO_RF24. Bug Fix gateway.py, change if is syntax at line 517. Update to update_code.py to fix issue when creating a new directory which contains no files but only sub-directories.
29/03/2022	1.7	290322	1.0	290322	Revised Settings Menu option, using a dropdown selection list, to improve performance. Update – All documentation referencing the Settings Menu. Bug Fix – updating Boost Settings.
30/03/2022	1.7	300322	1.0	290322	Improvement in graph load time.
31/03/2022	1.7	310322	1.0	290322	Further improvement in graph load times. Add waiting message when loading graphs.
01/04/22	1.7	010422	1.0	290322	Bug Fix – Daily usage graphs set x-axis to local time.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
08/04/2022	1.80	080422	1.0	030422	New Feature – Display graphs when Zone or standalone sensor tile is clicked. New Feature - A revised Home screen, which no longer auto-refreshes. The data and indicator fields on the screen are updated individually once every second. The result is that now any pop-up windows that are opened, will remain displayed until closed by the user. Update to Settings/Sensors Status to display additional information. WT32-ETH01 Gateway sketch – swap functions on pins 36 and 39 to match PCB configuration. Enable status indicators update on OneTouch page without page refresh.
09/04/2022	1.80	090422	1.0	030422	Weather data and Running Time not updating on page footer. Update to the Setting/System Status/Sensors popup to update the temperature values in real time.
10/04/2022	1.80	100422	1.0	030422	Bug Fix – Standalone sensors displayed before System Controller not updating. Update to make popup modal windows scrollable. Bug Fix – OneTouch Offset button not working.
11/04/2022	1.80	110422	1.0	030422	Bug Fix – Controller ‘last_seen’ datetime not updating on controller state change.
12/04/2022	1.80	120422	1.0	120422	Delay opening of Plotly graphs until modal window opens, to improve sizing on mobile devices. Update to the Home and OneTouch page refresh routines to allow the refresh period to be varied. A new Settings/System Maintenance/Page Refresh option has been added to allow the refresh period to be set between 1 and 15 seconds. Update to the database table ‘system’ to add a new field ‘page_refresh’.
13/04/2022	1.80	130422	1.0	120422	Home page refresh ‘hanging’ when trying to update fields where no data had changed.

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15/04/2022	1.80	150422	1.0	150422	All page refresh removed and replaced with page data updates at a configurable refresh rate. All Ajax data refresh PHP modules combined in a new PHP module 'ajax_fetch_data.php'. Zone popup disable schedule feature updated to enable toggling of the schedule enabled/disabled state. New column 'disabled' added to table 'schedule_daily_time_zone' to record enabled/disabled state. Bug Fix – schedule listing temperature circle not showing correct data.
16/04/2022	1.80	160422	1.0	150422	Bug Fix – Correct date/time positioning on schedule list page. Change schedule disabled colour to blue in zone popup.
17/04/2022	1.80	170422	1.0	150422	schedule list not displaying correctly and initial screen being loaded twice. Add disabled 'D' marker to schedule list where the schedule is either not enabled or all linked schedules have been disabled from the associated zone popups
19/04/2022	1.80	190422	1.0	150422	Bug Fix – schedule list not displaying correctly, re-apply #367 fix removed in error #368. Update to gateway.py for MQTT sensors, on update message_in queue on data change or every hour if no change. Update login.py banner script to Python3.
20/04/2022	1.80	200422	1.0	200422	Update to allow all sensor types to be configured to report either continuously or only on data change, with a variable timeout.
21/04/2022	1.80	210422	1.0	200422	Update to sensor reading timeout function, to use the last recorded sensor value as the current data value. Documentation update.
21/04/2022	1.80	210422_1	1.0	200422	Bug Fix gpio_ds18b20.py. Popup graphs display either Temperature or Humidity on y axix.
23/04/2022	1.80	220423	1.0	230422	Reimplement on-change feature for GPIO sensors. Update – add option to set the sensor resolution.

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23/04/2022	1.80	230422_1	1.0	230422	Cosmetic change to the Sensors GUI to hide field where applicable.
26/04/2022	1.80	260422	1.0	260422	Add an option to make individual user sessions persistent, rather than as currently they expire after 24 minutes of inactivity, default is non-persistent. Bug Fix to scheduling.php
28/04/2022	1.80	280422	1.0	260422	Update to the WT32-ETH01 Gateway sketch for when using NRF24L01 radio modules and correction to LED pin assignments.
02/05/2022	1.80	020522	1.0	260422	Bug Fix index.php to fix long standing error where trying to write lastlogin data to wrong table. Move WT32-ETH01 gateway sketch to nRF24L01 and RFM96 folders, with correct radio selection set. Update to refresh_session.php, to rewrite cookies. Bug Fix user_accounts.php - not saving enabled state when editing logged-on account.
03/05/2022	1.80	030522	1.0	260422	Replace header logged on user name with 'Session Timed Out' when a non-persistent session terminates.
04/05/2022	1.80	040522	1.0	040522	Further enhancement of the persistent session feature.
05/05/2022	1.80	050522	1.0	040522	Block user from applying code updates before applying any pending database updates.
12/05/2022	1.80	120522	1.0	040522	Improvement to disable/enable schedule within zone popup – update indicator and keep popup open until 'Close' button clicked.
16/05/2022	1.80	160522	1.0	160522	Bug Fix - Weather not showing temperature on header. Add flexibility to Repository selection data.
17/05/2022	1.80	160522	1.0	170522	Database change for future possible update
02/06/2022	1.80	020622	1.0	170522	Bug Fix – gateway.py not saving MQTT type humidity sensor values to zone_graphs table
09/06/2022	2.02	090622	1.0	060622	Major update from version 1.81 to version 2.02. Move to Bootstrap 5.2.1beta. Introduction of colour themes.
10/06/2022	2.02	100622	1.0	070622	Update Backup documentation

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11/06/2022	2.02	110622	1.0	110622	Add Burnt Orange Theme Add documentation for Themes Add documentation of Status Indicator Icon descriptions Change colour of Hysteresis status indicator to make bolder.
12/06/2022	2.02	120622	1.0	110622	Add missing logo for the Burnt Orange theme. Remove reference to Ionicons. Manual Backup will use Auto Backup destination if set or else will use /var/www/MySQL_Database/database_backups
13/06/2022	2.02	130622	1.0	110622	Bug Fix - start_backup.py to allow backups to networked folders.
14/06/2022	2.02	140622	1.0	110622	Bug Fix - start_backup.py resolve folder permission issue
15/06/2022	2.02	150622	1.0	110622	Bug Fix – boostlist.php, header colour
16/06/2022	2.02	160622	1.0	160622	Rename zone_graphs table to sensor_graphs. Delete all sensor_graphs table records older than 24 hours.
21/06/2022	2.02	210622	1.0	160622	Implement Database Restore menu option.
22/06/2022	2.02	220622	1.0	160622	Sort Documentation by Document Title.
24/06/2022	2.02	240622	1.0	160622	Add Sketch /nRF24L01/MaxAir_Temperature_Sensor_v035/MaxAir_Temperature_Sensor_v035.ino.
26/06/2022	2.02	260622	1.0	160622	Delete Arduino_Sketches/nRF24L01/PiHome_Temperature_Sensor_v034 Bug Fix – setup.php, creation of initial version of the ‘system’ table.
27/06/2022	2.02	270622	1.0	270622	Bug Fix – sensor_limits.php, set theme colours correctly. Bug Fix – reset ‘repository table to contain valid repository pointers.
28/06/2022	2.02	280622	1.0	280622	Bug Fix – chart_batteryusage.php, mask NULL bat_level values. Update to Auto Backup process to store last backup datetime in table ‘auto_backup’. This change to ensure that a backup is not triggered by a system restart.

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29/06/2022	2.02	290622	1.0	280622	Bug Fix – chart_addonusage.php, display correctly when node has multiple child ids, delete orphaned 'add_on_logs' start records before building graph data. Bug Fix – scheduling.php, set 'disabled' field in table 'schedule_daily_time_zone' table to be 0 by default.
30/06/2022	2.02	300622	1.0	280622	Update – gateway.py – only update 'nodes' table 'min_val' on first time see node, this will allow changes from GUI to persist. Bug Fix – model.php, Zone Modal text positioning.
03/07/2022	2.02	030722	1.0	280622	Bug Fix – mqtt_device.php, submit error when adding or editing 'MQTT Sensor' type devices. Bug Fix – change 'Senor' to 'Sensor' in all language files.
04/07/2022	2.02	040722	1.0	280622	Updated Bug Fix – mqtt_device.php, submit error when adding or editing 'MQTT Sensor' type devices.
11/07/2022	2.02	110722	1.0	280622	Bug Fix – Update any relay linked entries in the messages_out table when the relay node or child ids are edited
11/07/2022	2.02	110722_2	1.0	280622	Add documentation to describe message queue usage.
15/07/2022	2.03	150722_2	1.0	150722	New Feature – add negative sensor control slope zone type, uses a sensor and relay/s to control zone based on a negative sensor gradient. Update Zone Type categories, add new category 'Controller, Binary Sensor and NO System Control'. Update Zone Type categories, re-word the two existing 'Controller, Sensor and NO System Control' categories.
16/07/2022	2.03	160722_2	1.0	150722	Bug Fix – Switch type zones not displaying status correctly. Update Zone Type documentation
20/07/2022	2.03	200722	1.0	200722	Update – Zone GUI allows selection of Sensor Type. Update – Remove Zone Category 6, replaced by using Category 1 or 5 and a Binary Sensor. Update – Modify controller.php, homelist.php and ajax_fetch_data.php to use zone category 1 or 5, with Binary sensor type, in place of Category 6.

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20/07/2022	2.03	200722_2	1.0	200722	BUG FIX – controller.php extra ‘)’ in if statement at line 1387.
21/07/2022	2.03	210722	1.0	200722	BUG FIX – controller.php missing ‘;’ terminator at line 398.
22/07/2022	2.03	220722	1.0	220722	Update to http_messages table processing to use ‘zone_id’ rather than ‘zone_name’, to improve flexibility.
23/07/2022	2.03	230722	1.0	230722	Further update to http_messages table processing to remove redundant column ‘zone_name’. Bug Fix – zone.php, set footer colour correctly.
26/07/2022	2.04	260722	1.0	260722	New Feature – Implementation of ‘Message Sensors’.
05/08/2022	2.04	050822	1.0	260722	Bug Fix – add missing folder ‘/var/www/cron/logs’. Bug Fix – 1-wire sensor documentation.
07/08/2022	2.05	070822	1.0	070822	New Feature – allow creation of sensors to capture data from an eBUS compliant interfaced boiler
29/08/2022	2.05	290822	1.0	070822	Bug Fix – gateway.py. Update eBUS sensor documentation
06/09/2022	2.05	060922	1.0	070822	Update – Create and destroy relay messages_out entries with relay addition or deletion (rather than create with zone creation).
08/09/2022	2.05	080922	1.0	070822	Bug Fix – settings.php, remove redundant read of table ‘frost_protection’ (table no longer exixts). Bug Fix – controller.php, add timeout when reading Tasmota switch manual button using HTTP.
09/09/2022	2.05	090922	1.0	070822	compatibility changes for PHP version 8.x Update – add pin mapping for Orangepi 4 LTS
13/09/2022	2.05	130922	1.0	070822	Bug Fix – homelist.php and cron/controller.php – display target temperature correctly for category 1 type zones Update – install symlink if required for boards running Orange Pi OS to enable Adafruit Platform Detect to identify the board.
22/09/2022	2.05	220922	1.0	070822	Bug Fix – model.php, zone_controller table referenced instead of zone_relays.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
23/09/2022	2.05	230922	1.0	230922	Update – Add new field ‘n_id’ to messages_out table (future implementation of Gateway Relay Controller). Update – relay.php to reflect database change.
27/09/2022	2.05	270922	1.0	230922	Bug Fix – settingslist.php to capture the last CPU temperature reading correctly. Bug Fix – zone.php, not correctly maintain the page format when additional controller relays added. Bug Fix – js/request.js, not returning correctly when zones deleted. Update – gateway.py, apply trigger setting to MySensors relays when the sketch version is greater the 0.34 Update – gateway.py, in-coming message check for node-id changed from check for 0 (interger) to “0” (string) to match payload variable type. Update – gateway.py, relay updates use new messages_out field ‘n_id’ in place of ‘node_id’ to cope with instances where two node table entries may have the same node_id (GPIO controller and Gateway controller).
02/10/2022	2.05	021022	1.0	230922	Bug Fix – zone.php, not capturing sensor_type_id in Edit mode.
03/10/2022	2.05	031022	1.0	230922	Update – controller.php, functions.php to allow the controller script to execute with minimal zone settings.
08/10/2022	2.05	081022	1.0	230922	Bug Fix – functions.php – schedule start and end times require an offset for both timezone and daylight-saving time. Bug Fix – mode.php and homelist.php, when working in HVAC mode with button mode selected the lower right legend not being displayed correctly.
09/10/2022	2.05	091022	1.0	230922	Bug Fix – controller.php, move initialization of $system_controller array to outside of the main control loop.
10/10/2022	2.05	101022	1.0	230922	Bug Fix – mode.php, missing set variable $system_controller_id = $row['id'];.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
11/10/2022	2.06	111022	1.0	141022	Update – New option to allow Live Temperature control to be attached to any selected zone. Update – New option for Immersion and Cooling type zones to allow them to remain active and track to the zone's set default temperature when no schedule is active, until the next schedule becomes active.. Bug Fix – When the Mode is set to OFF all schedules and manual functions will be inoperative. Update – Remove table ebus_messages, replaced by sensor_messages.
18/10/2022	2.06	181022	1.0	141022	Bug Fix – gateway.py, when re-sync relays attached to the new combined gateway/controller, trigger not being applied. Update – WT32-ETH01 Gateway/Controller sketch.
20/10/2022	2.06	201022	1.0	201022	Update – Software Update processing – if a scheduler restart is required this is now handled within the script 'sw_install.py' rather than 'install.sh'.
21/10/2022	2.06	211022	1.0	201022	Enhancement – Optional Add-On software install of system image creation tools. New menu option to schedule creation of a system backup image file to a NAS drive. Bug Fix – sw_install.py use of space instead of tab. Bug Fix – model.php, initialise variable for software install status flag.
22/10/2022	2.06	221022	1.0	201022	Update – image-backup utility, use wildcard when checking root volume type for Armbian OS compatibility. Bug Fix – restore ebus_messages table.
22/10/2022	2.06	221022_2	1.0	201022	Only allow installation of Image Tools on systems running Raspian.
23/10/2022	2.06	231022	1.0	201022	Bug Fix – check_gw.php trying to retrieve process info when no process running.
26/10/2022	2.06	261022	1.0	201022	Bug Fix – zone.php, not updating the zone_id field in the messages_out table.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
02/11/2022	2.06	021122	1.0	201022	Bug Fix – HA_HVAC_integration.py, when none RPi boards do not process under voltage. Bug Fix – Home Assistant install requirements file requirements.txt. Documentation – 'Setup Guide Home Assistant Ingegration', add case where Mosquitto broker is already installed. Documentation – update ReadMe for WT32-ETH01 Gateway version 2.
06/11/2022	2.06	061122	1.0	061122	Bug Fix – In some instances the message_out table zone_id field is not populated correctly, this database update refreshes the data.
12/11/2022	2.06	121122	1.0	061122	bug: fix launching find_mygw instead of gateway
27/11/2022	2.06	271122	1.0	061122	Bug Fix - gpio_ds18b20.py, only process sensors which already have a database entry, code aligned with gateway.py.
27/11/2022	2.06	271122_2	1.0	061122	Update – holiday.php and holidayslist.php to apply theme colours correctly.
01/12/2022	2.06	011222	1.0	061122	Update – Add documentation for Holidays function. Bug Fix - relay.php and Arduino sketch for Gateway Controller, use distinct type id of 25.
07/12/2022	2.06	071222	1.0	061122	Update – sketch MaxAir_GatewayWT32_ETH01_v002.ino. Bug Fix - zone.php maintain default value not being captured or displayed correctly.
10/12/2022	2.06	101222	1.0	101222	Update – add new table 'relay_logs' to capture state changes sent to the relays. Update – gateway.py populate table 'relay_logs' whenever a state change is applied to a relay. Update – db_cleanup.php clear new table 'relay_logs' after a set interval.
13/12/2022	2.06	131222	1.0	131222	Update – new option in Settings/Device Configuration/Test Relays, allows the state of any 'created' relay to be toggled OFF/ON (System Controller operation is suspended whilst dialogue is active).

Date	Ver	Build	DB Ver	DB Build	ChangeLog
19/12/2022	2.06	191222	1.0	131222	Update – ebus.py, implement 'Mode', 'Timeout', 'Resolution' and 'Correction Factor' for EBus type sensors. Update – controller.php, flag any zones that have missing sensors or relays and continue processing. Bug Fix – db.php, zone_delete function using wrong table name.
22/12/2022	2.06	221222	1.0	211222	Update – move_files.py, force restart of ebus.py if file updated. Bug Fix – change table 'system_control;er', field 'max_running_time' from type TINYINT to UNSIGNED TINYINT to match GUI maximum value of 180.
28/12/2022	2.06	281222	1.0	211222	Update – gateway.py improved error trapping.
29/12/2022	2.06	291222	1.0	211222	Bug Fix – db.php – processing add_on switch zone relays - $category variable not set. Bug Fix – missing sub-mode 8 added for max_running_time exceeded.
31/12/2022	2.06	311222	1.0	211222	Update – Enable STATE topic generation for MQTT Controller devices.
03/01/2023	2.06	030123	1.0	211222	Update - Enable STATE topic generation for MQTT Controller devices, where multiple relays are attached to the same MQTT controller.
11/01/2023	2.06	110123	1.0	211222	Bug Fix – gpio_ds18b20.py – remove test code left in place.
21/01/2023	2.07	210123	1.0	210123	New Feature – allow a delay time in seconds to be applied to any relay ON command.
23/01/2023	2.07	230123	1.0	210123	Update – Relays documentation for Lag Time. Update – gateway.py – suppress outgoing messages when relays are waiting for lag delay to expire.
24/01/2023	2.07	240123	1.0	210123	Bug Fix – gateway.py, relay initialization now used any lag delay settings.
25/01/2023	2.07	250123	1.0	210123	Update/Bug Fix – gateway.py, process all relays where 'sent'=0 in a single pass, rather than 1 relay per pass.
27/01/2023	2.07	270123	1.0	210123	Update – Pin_Dict.py, add GPIO support for OrangePi 5.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
30/01/2023	2.07	300123	1.0	210123	Update – PHP script ‘apply_databse_updates.php’ added to ‘MySQL_Database’ directory, used to apply missing database updates. Update – example databases – gateway changed from serial to virtual.
17/02/2023	2.07	170223	1.0	210123	Update – gateway.py to apply sensor correction factor for MQTT sensors. Update – check_gw.php cosmetic change to display text. Update – add new repository for Version 3.
23/02/2023	2.07	230223	1.0	210123	Bug Fix – onetouch.php – footer running time query using wrong table name and requires setting of system_controller_id variable.
25/02/2023	2.07	250223	1.0	210123	Update - onetouch.php to displav information message if LiveTemp is selected for a zone which is not in an active state.
26/02/2023	2.07	260223	1.0	210123	Update – homelist.php, db.php and request.js, improvement when clicking on an add-on zone tile to toggle the zone ON/OFF state.
28/02/2023	2.07	280223	1.0	210123	Bug Fix – db.php Add New Job not closing dialogue window.
01/03/2023	2.07	010323	1.0	010323	Update – add new column ‘status-prev’ to table ‘zone_current_state’. Used to improve operation od ‘switch’ type zones when state is changed by clicking on the zone tile. Associated changes to ‘controller.php’ and ‘gateway.py’.
05/03/2023	2.07	050323	1.0	010323	Bug Fix – controller.php, syntax error in query definition for “INSERT INTO zone_current_state …….
09/03/2023	2.07	090323	1.0	090323	Update – controller.php, db.php, model.php and request.js to improve operation of the ‘Test Relays feature. Update – Add documentation for combined Gateway/Controller, with Relay Adapter board.
10/03/2023	2.07	100323	1.0	090323	Update – gateway.py, bypass any set relay lag time when using the ‘Test Relays’ option.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
24/03/2023	3.00	240323	1.0	230323	Update – Implement Version 3.00. Main change is to replace ‘controller.php’ with ‘controller.py’. ‘controller.py’ is a continuously running script rather than a scheduled task, this improves overall responsiveness of the GUI. A new watchdog job ‘check_sc.php’ is implemented and added to the job scheduler to restart the new ‘controller.py’ script. Update – changes to scheduling so that the current schedule state and its ‘id’ are stored in the ‘zone_current_state’ table, where updates are implemented in the ‘controller.py’ script, function ‘get_schedule_status’ in ‘st_inc/functions.php’ is no longer used and is replaced by a function within script ‘ccontroller.py’. Update –sensor current values are stored on update within 2 new fields in table ‘sensors’, the Home screen GUI retrieves these values, rather than reading the ‘messages_in’ table. Update – for the Gateway/Controller hardware the ‘heartbeat timer’ timeout value is set in the Gateway GUI. Update –update example databases to Build 210323. Update –create new Settings option ‘Settings/System Status/Scripts Status’ to display information relating to the ‘gateway.py’ and ‘controller.py’ scripts. Bug Fixes – shutdown and reboot process changed to check flags in the ‘system’ table, new scheduler job added to control script ‘shutdown_reboot.py’. Bug Fixes – various fixes to enable the ‘gateway.py’ and ‘controller.py’ scripts to run without error where the system has not yet been fully configured.

Date	Ver	Build	DB	DB	ChangeLog

			Ver	Build	
24/03/2023	3.00	240323	1.0	230323	Update to Bootstrap Version 5.2.3 Update to Bootstrap Icons Version 10.1.3
31/03/2023	3.00	310323	1.0	310323	Bug Fix – Add On zones do toggle from the GUI, fix to request.js and db.php. Tidy Up – ajax_fetch_data.php. use existing variable in place of table reference. Update – new field 'test_run+time' added to 'system' table, to be used in future release to run MaxAir at an artificial time as a way to test scheduling. Update – zone and boiler controller NRF24L01 sketches and gateway.py to add 'heartbeat' feature.
01/04/2023	3.00	010423	1.0	310323	New Feature – A new option Settings/System Configuration/Schedule Test allows you to select a date and time to be used by MaxAir as its current time, MaxAir will move to this date and time and then virtual time will progress forwards from this start point. The time displayed on the toolbar will be annotated to indicate that the system is operating in a TEST MODE. Update – Icon changes, change MQTT Settings button to display icon, change System Controller Settings dropdown and button icon.
05/04/2023	3.00	050423	1.0	310323	Bug Fix – Auto Backup when using a 1 DAY frequency. Update – update to various button icons. Bug Fix – controller.py not capturing schedule count per zone correctly.
06/04/2023	3.00	060423	1.0	310323	Update – apply schedule and test mode run time updates to deprecated 'controller.php', which is kept as a 'fallback' option.
09/04/2023	3.00	090423	1.0	310323	Bug Fix – auto_backup.py fix for 1 DAY frequency. Bug Fix – controller.py, set time_id = 0 when no schedule is found. Update – gateway.py, use dictionary to hold heartbeat timers for multiple relay controller.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
10/04/2023	3.00	100423	1.0	310323	Bug Fix – gateway.py, convert MQTT payload to float type for cases where no ‘Attribute’ is used.
12/04/2023	3.00	120423	1.0	310323	Update – Allow custom tile messages for Binary Sensors, rather the simply ‘ON’ or ‘OFF’.
16/04/2023	3.00	160423	1.0	310323	Bug Fix - controller.py, weather factoring when using a user sensor
04/09/2023	3.00	040923	1.0	040923	Update – Added a new option ‘Settings/Device Configuration/Hide Sel/Rel’. Administrators can select to hide stand-alone Sensors and Add-On Zones on the Home screen for non-admin users.
08/09/2023	3.00	080923	1.0	040923	Update – Allow unallocated sensor to be associated with a SWITCH type zone, so that the sensor tile is displayed adjacent to the add-on zone tile.
21/09/2023	3.00	210923	1.0	040923	Update – gateway.py- improvement in processing of MQTT sensors. Update - New option Settings/Device Configuration/Hide Sen/Rel to allow Sensors and/or Relays to be hiden on the Home Screen for a particular user.
25/09/2023	3.00	250923	1.0	250923	Further Update – gateway.py- improvement in processing of MQTT sensors. Update – table ‘nodes_battery’, change field ‘node_id’ from int(11) to char(50), to match ‘nodes’ table definition.

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08/10/2023	3.00	081023	1.0	071023	New Feature – gateway.py, notice.py, model.php, mqtt_device.php and database update 260923.sql, to enable the setting of 'Notice Interval' and 'Minimum Value' on the MQTT Devices screen. Update – further updates to MQTT device handling to enable Battery graphs and Sensor/Relay loss of communication indications. Bug Fix – table 'http_messages' correct 'node_id' field format. Bug Fix – model.php, db.php and request.js – selection of sensors to graph query correction. Bug Fix – homelist.php – 'Add-On' type zones to only display a single tile, even when multiple controllers are attached. Bug Fix - model.php – fix error where 'unallocated' sensors are not detected correctly. Bug Fix – model.php – fix error where 'unallocated' type 1 relays are not detected correctly.
27/10/2023	3.00	271023	1.0	071023	Update – create local Python module 'schedule.py' and remove check in '/var/www/cron/install_jobs.sh'. Update –Image Tools.
30/10/2023	3.00	301023	1.0	071023	Update – nmcli_reboot_wifi.sh – retrieve connection name using the 'nmcli con' command.
07/11/2023	3.00	071123	1.0	071023	Bug Fix – setup.php – creation of initial gateway record not adding a default value for field 'heartbeat_timeout'.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
10/11/2023	3.00	101123	1.0	071023	Update – gateway.py – Python module 'telnetlib' (used to send and receive TCP data packets) is deprecated in Python version 3.11 and will be removed in version 3.13. Replaced with socket.sendall and socket.recv(1). As there is no direct replacement for telnetlib.readline(), a separate program thread has been created to read the socket stream until a newline character has been detected and compose a message queue, which is read line by line by the main program thread. Update – gateway.py – a logfile to monitor incoming 'heartbeat' messages is created at '/var/www/logs/gateway_heartbeat.log'. The logfile is automatically emptied at midnight each day.
11/11/2023	3.00	111123	1.0	071023	Bug Fix – gateway.py – fix Queue error handling for 'Empty' Exception. Update - Image Tools install.sh and image filename format generated by auto_image.py
12/11/2023	3.00	121123	1.0	071023	Bug Fix – setup.php – remove update to heartbeat_timeout field in table gateway, it will be set by database update 170323.sql Update – model.php – change to ethernet model to accommodate different ethernet interface names (eg. eth0, end0).
14/11/2023	3.00	141123	1.0	071023	Bug Fix – gateway.py – correct Queue syntax for 'Empty' exception.
23/11/2023	3.00	231123	1.0	071023	Update – gateway.py, sensor.php – ensure that when a sensor is operating in 'On-Change' mode that the 'Timeout' value is a minimum of 10 minutes and that 'gateway.py' will ignore any zero 'Timeout' values.
24/11/2023	3.00	241123	1.0	071023	Update – model.php, sensor.php – further refinement to Sensors GUI to correctly process changes between Continuous and On_Change modes, also add icon for On-Change mode.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
30/11/2023	3.01	301123	1.0	071023	Update – Bootstrap Icons update from 1.10.3 to 1.11.0 Bug Fix – model.php, syntax changed for ethernet interface name check in Ethernet Modal. Update –update Bootstrap from version 5.2.3 to 5.3.2
05/12/2023	3.01	051223	1.0	071023	Update – gateway.py – trap deadlock errors when updating nodes table to resolve gateway reboot during Auto Backup.
01/01/2024	3.01	010124	1.0	010124	New Feature – by default sensor readings are written to the database 'messages_in' table, this is now selectable from the Settings/Device Configuration/Sensors dialogue For sensors where there is no requirement to graph readings, this will reduce the size of the database.
02/01/2024	3.01	020124	1.0	010124	Bug Fix – add missing updates for gateway.pt and ebus.py for commit #497.
11/01/2024	3.01	110124	1.0	010124	Bug Fix – controller.py – zone_controller_logs table not capturing zone switch off state correctly. New Feature – Settings/System Status menu item SC/Zone Logs popup added to display the current log entries for both the System Controller and he zones. The popup is dynamic, hence will reflect changes to the 'controller_zone_logs' table. Update – 'controller_zone_logs' examples added to the 'Basic Fault Finding' document. Update – Convert Setting/System Status/Uptime to a dynamic updating modal.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
14/01/2024	3.01	140124	1.0	010124	Update – controller.py – add new ‘start_cause’ ‘Restart Schedule’, to indicate continuation of as current schedule, after a temporary stop. Update – Convert Home screen System Controller popup to a dynamic updating modal. Bug Fix – gateway.py – not saving readings to messages_in table when set to not graph and NOT disable database saving.

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07/02/2024	3.02	070224	1.0	070224	Update – All code for Settings screen popup modals moved to module 'model.php'. Update – Improvements to update of Settings screens dynamic data display. Bug Fix – Settings/System Configuration/Themes was not showing confirmation request when 'Delete' selected. Bug Fix – 'Setup Holidays Schedule' document – clarification of the operation of the 'Holidays' Home screen button. Bug Fix – Holidays Scheduling the 'Enable' checkbox flag not being saved to table 'holidays. Bug Fix – Settings/System Status/Job Status – only allow selection of 'enabled' jobs. Update – New Settings/System Status menu option entitled 'Zone State'. This will dynamically display the current state of the individual zones. Bug Fix – 'db_cleanup.php' – when deleting 'gateway_logs' table entries older than set interval, leave the latest record in place. Bug Fix – 'controller.py' – Pump type relays not being processed. Bug Fix – 'gateway.py' – 'relay_logs' table entries not being generated for add_on type switches. This change includes the addition of a new column, 'add_on_toggle' to the 'zone_current_state' table. Bug Fix – 'controller.py', 'db.php' and 'binarySet.php' – 'add_on_logs' table entries not being generated for add_on type switches. Update – 'controller.py' – New Settings/System Status menu option entitled 'Relays'. This will dynamically display the current ON/OFF state of the individual relays, clicking on the status indicator will display the historical state changes. Update – 'install.sh' for HomeKit Add-On.

Date	Ver	Build	DB Ver	DB Build	ChangeLog

10/02/2024	3.02	100224	1.0	070224	Bug Fix – 'gateway.py', not resetting relay switch on lag time after first use. Update - Settings/System Status/Relays, will now display the status column in orange if the relay is waiting for switch on lag time to expire. Bug Fix – 'gateway.py', sensor 'on-change' mode setting not being applied to the 'sensor_graphs' table.
11/02/2024	3.02	110224	1.0	070224	Bug Fix – 'model.php'. enable selection of 'max child_id when creating new MQTT type nodes.
19/02/2024	3.03	190224	1.0	150224	Update – 'index.php', improve logon time by minimising calls to 'userhistory' table. Update – 'gateway.py', improvement in messages in queue handling, such that the queue is emptied on a single cycle of gateway.py. Update – optionally archive sensor readings to a 'CSV' file. Update – optional new graph page to display minimum and maximum sensor readings, based on the data in the new 'CSV' file. Update – new schedule task 'csv', to execute the script 'var/www/cron/graph_csv.py' at midnight each day. This task will optionally write the CSV file to a location set using Settings/Systems Configuration/Archive Graphs'.
21/02/2024	3.03	210224	1.0	150224	Bug Fix – 'index.php', #505 broke 'persistent' user feature. Update – 'Settings/Node and Zone Configuration' allow editing of the 'Max Number of Child IDs' field.
23/02/2024	3.03	230224	1.0	150224	Bug Fix – 'gateway.py', not updating 'sensor_graph' table for Switch type zones with an associated sensor.
25/02/2024	3.03	250224	1.0	250224	Update – Hide 'One Touch' and 'Mode' tiles for nonadministrator. Add 'Live Temp' tile to the 'Set Buttons' option, to allow display for nonadministrator users.
27/02/2024	3.03	270224	1.0	250224	Update – Add documentation for User Account setup and user Home screen configuration.

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03/03/2024	3.03	030324	1.0	250224	Update – 'gateway.py' – install python module 'pymysql' in preparation for a future change from using 'MySQLdb'.

					Bug Fix – ‘weather_update.php’ – missing field ‘enable_archive’ in SQL query to add first record the ‘weather’ table.
05/03/2024	3.03	050324	1.0	250224	Update – ‘functions.php’, add new function ‘os_info()’ to capture OS information, use to add OS Distribution to Settings/System Status/OS Version popup. Bug Fix – ‘controller.py’ – fix for schedule Offset, not capturing required parameters.
06/03/2024	3.03	060324	1.0	250224	Update – Enable detection of Linux Distribution and select WEB user id (www-data or http) dependant on distribution, include in setup and add-on configuration scripts. Bug Fix – Remove page errors when MaxAir is opened with no configuration. Bug Fix – Offset tile not updating the Offset Value.
09/03/2024	3.03	090324	1.0	250224	Add sketch for Temperature Sensor PCB, which uses jumpers to set the Node ID. Bug Fix – Suspend gateway processing whilst the script ‘db_cleanup.php’ is executing to prevent ‘gateway.py’ erroring due to concurrent table access requests.
12/03/2024	3.03	120324	1.0	250224	Bug Fix – ‘setup.php’ not detecting OS version correctly.
13/03/2024	3.03	130324	1.0	250224	Update – ‘update_code.py’ enable for ArchLinux user ‘http’.
23/03/2024	3.03	230324	1.0	250224	Update – ‘Image Tools’ to latest version. Update – ‘auto_image.py’ to work with OrangePi boards.

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15/04/2024	3.03	150424	1.0	250224	Update – ‘auto-image.py’ to stop ‘pihome_jobs_schedule.service’ in order to suspend database updates while the image creation task is running, to prevent possible MySQL corruption. Bug Fix – ‘update_code.py’ not setting WEB userid correctly for Ubuntu os. Bug Fix – ‘gateway.py’ capture relay state changes to ‘relay_logs’ table for Pump type relays attached to multiple zones.
22/04/2024	3.03	220424	1.0	250224	Bug Fix – ‘update_code.py’ not setting WEB userid correctly for Raspian os. Update – PCB design of Gateway ESP8266 to ESP32 Adapter. Update – Sketch for ESP32 Gateway Adapter.

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26/07/2024	3.04	260724	1.0	260724	Bug Fix – 'Away' function not working correctly for 'add-on' type zones. Update – 'Away' function will now restore relay states as they were before 'Away' was activated. Update – PCB design for 8way relay controller. Update – Sketch added for 8way relay controller, uses 'heartbeat' signals to avoid the need to continually rewrite ON/OFF values. Update – New sketch added for the PiHome Zone Controller, uses 'heartbeat' signals to avoid the need to continually rewrite ON/OFF values. Update – New column 'state' added to table 'relays', used by 'gateway.py' and 'controller.py' to synchronise MySensor relay adapters which implement the 'heartbeat' signals. Update – 'gateway.py', new message sent to all MySensor relay adapters which implement the 'heartbeat' signals, this message contains a mask for any unallocated relays in order that the adapter sketch can clear any unallocated relays. Update – 'model.php', change to detect if 'wlan0' interface is present and display appropriate message if not. Commit #519 – New Feature – 'Settings/System Configuration/Zone State Log' option implemented to allow log files to be created to show changes in the 'zone_current_state' table. Individual files created for each zone in directory '/var/www/logs', the file name format is 'zoneXX_current_state.log' where XX is the zone ID. New fields added to the 'zone_current_state' table. Bug Fix – 'controller.py', change to the creation of 'zone_current_state' table entries to improve the logging of zone relay state changes

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07/03/2025	3.05	070325	1.0	300824	Bug Fix – Roll-Up various updates. Update – Sketch for WT32-ETH01 combined gateway and relay controller, uses ‘heartbeat’ signals to avoid the need to continually rewrite ON/OFF values and a ‘mask’ for the current relays state to be used to ensure that the relays remain synchronized. Additional change to display the currently connected Node IDs on the gateway WEB interface. Update – ‘gateway.py’ and ‘controller.py’ where MySensor relays are connected using a WT32-ETH01 combined gateway and relay controller, relays will only be written to when a state change occurs. Database Update – New column ‘boost_button_state’ added to table ‘boost’.
13/03/25	3.06	130325	1.0	300824	Update – 'gateway.py', 'controller.py', 'Maxair_Multi_Controller_v034.ino' and 'Maxair_Multi_Controller_v034.ino'. Change so that v034 works in the same manner as the PiHome Zone Controller, where the controller is written to continuously. Sketch v035 is a development version, where the controller is only written to on state change. Update - 'Change Log' now save as an 'odt' type file rather than a 'docx'. Update – 'MaxAir_GatewayESP32.ino' – added function to reconnect to router WiFi connection, on loss of signal, retry count added to WEB console. Update – Added documentation for configuration jumper setting for Sensor PCB Ver 2.12 and Multi Controller PCB Ver 1.0. Update – 'gateway.py' – change to check of sketch version to catch non-numeric values. Update – 'HA_HVAC_integration.py' remove dependancy for 'paho-mqtt' version 1.5.0 and modify 'install.sh' accordingly.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
07/07/2025	3.07	070725	1.00	100725	Update – Bootstrap Version 5.3.7, requires a browser cache refresh. Database Update – 'zone_view' – align zone_sensor and zone_controllers JOINS. Database Update – 'sensors'. 'relays' – added columns 'fail_timeout' and 'last_seen'. Update – 'zone.php', 'homelist.php', 'ajax.php', ajax_fetch_data.php', 'footer.php', 'sensor.php', 'relay.php', 'notice.py' and 'controller.py' – allow multiple sensors to be attached to a zone, with the control temperature being the average of the sensors attached to the zone. Both individual sensors and relays can have timeout set, to indicate that they have failed to report their status within a time limit.
20/07/2025	3.08	200725	1.00	100725	Bug Fix – 'homelist.php' – fix to zone sensor query which was stopping the Nome page opening Update – 'notice.py' – add 'Message Sent' timestamp and tag each status message with its timestamp..
29/07/2025	3.09	290725	1.00	100725	Bug Fix – 'gateway.py' – fix to ensure that if a system power failure occurs during a zone overrun phase, then the system will return with the zone relay switched OFF. Update – 'settingslist.php', 'model.php', 'zone.php', 'sensor.php' and 'relay.php' – if zone, or sensor, or relay edit is initiated from the Settings menu, then after committing the change, the user is returned to the popup menu from which the edit was initiated.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
04/07/2025	3.10	040825	1.00	020825	Database Update – '020825.sql' – new table 'sensor_average', used to define a multiple sensor device, initially populated using any zones which have multiple sensors attached. Update – 'controller.py', 'model.php', 'zone.php', 'db.php', 'chart.php', 'chart_load.php' and 'chart_min_max.php' – 'controller.py' used to generate 'message_in' and 'sensor_graphs' table date for multiple sensor devices, graphs page able to display Average Sensor graphs and Settings/System Configuration/Select Graphs popup able to display Average Sensor selections
18/08/2025	3.11	180825	1.00	070825	Database Update – '070825.sql' – change table 'user', rename field 'admin_account' to 'access_level'. Update 'access_level' = 1 to 'access_level' = 0; else 'access_level' = 1. Update – 'index.php, 'header.php', 'homelist.php', 'mode.php', model.php', 'onteouch.php', 'refresh_session.php, 'settings.php', 'setup.php' and 'user_accounts.php' used to control the level of access individual users have to the system. Update – 'gateway.py, 'relay.php', 'zone.php', 'footer.php' and ‘db.php' – add support for ‘Olimex’ combined gateway, relay controller and sensor. Improve zone deletion.
19/08/25	3.11	190825	1.00	070825	Bug Fix – 'index.php’ – page error when username/password is incorrect.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
20/08/25	3.11	200825	1.00	070825	Update – 'homelist.php' and 'ajax_fetch_data.php' to update the 'Mode' tile on change. Add Feature – 'modeSet.php' as an API to set the System controller operating mode.
10/09/25	3.11	100925	1.00	070825	Bug Fix – 'gateway.py' – reading MQTT Sensors – test if the message format is either a float variable or a python dictionary. Bug Fix – 'setup.php' – revert to previous version. Update – new table 'frost_sensor_relays', used to link sensors used for frost control to zone relays. Update – 'sensors.php', 'footer.php', 'model.php, 'ajax_fetch_data.php' and 'controller.py' – changes to 'Frost Control' to allow multiple zone controller relays to be attached to a single sensor being used for frost control. Add Feature – New software install Add On to deploy Mosquitto Broker and Zigbee2MQTT. Change to 'gateway.py' for zigbee2mqtt message.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
27/11/25	3.13	271125	1.00	271125	Update – new table 'sensor_min_max_graph', used to hold the daily minimum and maximum vales for each sensor set to produce a graph. Update – 'chart_load.php', 'footer.php', 'chart_min_max.php, and 'graph_csv.py' – changes to production and display of Min/Max sensor graphs. Date now stored in a table rather than a 'csv' file. It was found that the graph processing routine could not cope with very large 'csv' files. Min/Max table entries are added for each sensor on a daily schedule. Note: the previous process for producing the 'csv' file remains as it was.
29/11/25	3.14	291125	1.0	291125	New Feature – new column 'one_shot' added to 'boost' table', used to hold a flag to indicate that boost will terminate when set temperature reached. New Feature – 'boostlist.php', 'model.php', 'db.php', 'request.js' and 'controller.py' – implementation of 'One Shot' Boost feature. Boost can optionally be set to finish when the target temperature is reached, if the target is never reached, then boost finishes after set interval.
29/11/25	3.15	291125	1.0	010126	Bug Fix – increase field size of column 'zone_mode' in 'relay_logs' table' from char(50) to char(70).

Date	Ver	Build	DB Ver	DB Build	ChangeLog
06/01/26	3.16	060126	1.0	060126	New Feature – Database update to table 'relays', add columns 'index_is', 'pre_post', 'show_it' and 'message_in'. Adds support for stand-alone relays and Home Screen positioning of relays. New Feature – Stand-alone relays, enable relays not associated to any zone, to be displayed on the Home Screen. The relay, on/off state can be toggled by clicking on the stand-alone relay tile. Changes to files – 'relay.php', 'ajax_fetch_data.php'. New Feature – Zone, Add-On Zone, Stand-Alone Sensors and Stand-Alone Relays, Home Screen tile placement. New GUIs 'Settings/Node and Zone Configuration/Zone Indexing' and 'Settings/Device Configuration/Device Indexing'. Changes to files 'settings.php, 'settingslist.php', 'ajax_fetch_data.php', 'modal.php', 'homelist.php', 'db.php', 'request.js', associated documentation files and all language .php files. Bug Fix – 'gateway.py' trap MySQL error 1020 when attempting to delete aged records from 'sensor_graphs' table (caused by multiple threads trying to update concurrently).
07/01/26	3.17	070126	1.0	060126	Update – Bootstrap Version 5.3.8, requires a browser cache refresh.
08/01/26	3.17	080126	1.0	060126	Update – 'model.php' – simplification of the 'Settings/System Status/Maxair versions' pop-up..

Date	Ver	Build	DB Ver	DB Build	ChangeLog
17/01/26	3.17	170126	1.0	060126	Bug Fix – 'gateway.py' – MQTT relays were not disabled when 'Enable Outgoing Messages' was de-selected in the Gateway Setup GUI. Bug Fix – 'gateway.py' – updating 'nodes' table 'last_seen' when processing MQTT sensors caused a database update error. Resolved by placing the update of the 'nodes' table after the update of the 'mqtt_devices' table.
18/01/26	3.17	170126	1.0	060126	Bug Fix – 'model.php' – incorrect code functions deployed for Zone and Device Indexing.
23/01/26	3.17	230126	1.0	060126	Update – 'model.php' – increase size of 'Jobs' pop-up. Update – 'Image_Tools' Add-On Bug Fix – 'sensor.php' – table 'sensors' column 'frost_controller' is now redundant, so always set to 0.
29/01/26	3.18	290126	1.0	060126	Update – all language.php files – add ability to maintain an exceptions language file, which will be applied as an overlay, enabling users to customise label text, that will not be overwritten by the update process.
03/02/26	3.18	030226	1.0	060126	Bug Fix – 'model.php' – Relays pop-up 'Show' checkbox disabled when it should be enabled.
06/02/26	3.18	060226	1.0	060126	Update – 'prerequisites/install.sh' – make trixie compliant, optional install of 'emsctl' and 'log2ram'. Update – 'setup.php' = add jobs 'check_ebus.php' and 'check_ens.php' and to set the default Repository to 'twa127/PiHomeHVAC'. Update – add new script 'check_ems.php'.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
11/02/26	3.18	110226	1.0	060126	Update – 'gpio_ds18b20.py' can be passed a flag to disable smoothing and exponential weighted moving average. If no argument is used, then the feature is enabled, if a '1' argument is passed, then the feature is disabled eg. 'python gpio_ds18b20.py 1". New Feature – 'check_ds18b20_1.php", alternative 'watchdog' script for python script 'gpio_ds18b20.py', will pass a 1 argument to disable smoothing and exponential weighted moving average. Update – 'gpio_ds18b20.py' modified to make it compatible with systems where the 1-wire interface is configured to operate on more than one GPIO pin.
14/02/26	3.18	140226	1.0	060126	Update – 'gpio_ds18b20.py' test for a valid response enhanced to check for 'crc=00"'. Update – 'notice.py' change to message text for sensor timeout failure.
18/02/26	3.19	180226	1.0	180226	New Feature – Database update to table 'schedule_daily_time', add column 'smart_off'. Adds support for 'smart off' feature. New Feature – 'schedulelist.php', 'scheduling.php', 'functions.php', 'controller.py' and language files - Zone state will not re-activate within a set number of minutes of a schedule end time. Courtesy

Date	Ver	Build	DB Ver	DB Build	ChangeLog
19/02/26	3.20	190226	1.0	190226	New Feature – Database new table 'relay_messages' and update to table 'relays', add column 'current_val_2'. Adds support for 'relay messages' feature. New Feature – 'settingslist.php', 'model.php', 'db.php', 'relay.php, 'ajax_fetch_data.php', 'request.js' 'rc10_control.py' and language files – Enable custom message text to be displayed in the lower right hand corner of Stand Alone Relay tiles.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
04/03/26	3.21	040326	1.0	040326	New Feature – new columns group_id', 'prev_schedule', 'schedule' and 'sch_time_id' added to 'relays' table', new table `schedule_daily_time_relays`, new table 'relay_group', new view `schedule_daily_time_relays` - used for relay scheduling. New Feature – Stand Alone Relay Scheduling – 'schedulelist.php', 'scheduling.php', 'ajax_fetch_data.php'. 'homelist.php', relay.php, 'gateway.py' 'controller.py' and language files. – implementation of relay activation scheduling for stand alone relays. New Feature – Stand Alone Relay Grouping – 'homelist.php, 'settingslist.php, 'model.php' and 'ajax.php' – used to enable relays to be grouped together and actions applied to all relays within the group. Update – stand alone relay actions added to relay logs New Feature – MQTT Relays – 'mqtt_device.php', 'controller.py' – added support for some Shelly and SonOff relays which are capable of reporting the current switch ON/OFF state. MaxAir will use this reported state to recover from MQTT relay power failure. New selection field 'Brand' added to the MQTT Devices setup screen to allow selection of either Tasmota, Shelly, Sonoff or Other. Bug Fix – 'model.php' – when updating Email Setup the already saved encrypted password stored in the database is encrypted again. Update – 'model.php' – added ability to display the password field in plain text.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
11/03/26	3.21	110326	1.0	040326	Bug Fix – 'gateway.py' – capture of MQTT state messages for dual relays not referencing first relay correctly. Bug Fix – 'controller.py' – synchronisation of MQTT Pump relays incorrect.
31/03/26	3.22	310326	1.0	310326	New Feature – 'Hot Water Compensation' – new table 'hw_compensation', new column 'smart_off' in table 'schedule_daily_time', 'controller.py', 'homelist.php', 'settingslist.php', 'ajax_fetch_data.php', 'db.php', 'request.js' and language files – calculate the target hot water temperature based on a 'control' temperature and a set of defined coefficients. New Feature – 'Show Disabled Running Schedule Icon' – 'controller.py', 'homelist.php' and 'ajax_fetch_data.php' – display a white X in a blue circle icon in the lower right hand corner of a zone tile when a schedule which should be currently running, has been disabled.
01/04/26	3.22	010426	1.0	010426	Update – 'Disabled Schedule Icon' – new column 'show_disabled' in table 'schedule_daily_time', 'controller.py' and 'scheduling.php' – for Zone type schedules tick box option to enable/disable the display a white X in a blue circle icon in the lower right hand corner of a zone tile when a schedule which should be currently running, has been disabled. The default is enabled.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
05/05/26	3.23	050526	1.0	050526	New Feature – 'Return to Home Page' – new column 'page_timeout' in table 'system', 'settings.php', 'schedule.php' , 'onetouch.php', 'db.php', 'model.php', 'request.js' and language files – if the system is left displaying any Settings menu screen, the Schedule list screen or the One Touch screen for longer than a configurable timeout period, then the display will return to the Home Screen. The timeout can be configured from the 'Settings/system Maintenance/Page Refresh' menu option, selecting a value of '0' will disable the timeout. The default is 60 second timeout. New Feature – 'Temperature Delta' – “settings.php', settingslist.php', 'model.php', 'ajax_fetch_data.php and language files – new menu option 'Settings/System Status/Temp Delta', to display the Max, Min and Delta values by either Zone or Sensor over the preceding one or twenty four hour time interval. New Feature – 'Disable schedules during summer months' – new table 'summer', new column 'disable_in_summer in table 'schedule_daily_time', 'controller.py', 'homelist.php', 'onetouch.php'. 'scheduling.php', 'model.php', 'settingslist.php', 'ajax_fetch_data.php', 'db.php', 'request.js', 'functions.php' and language files – disable selected schedules during a set period of time, popup to define the period and a 'OneTouch' button to enable/disable the feature. New Feature – New APIs – modeSet.php, getZoneData.php, getFrostStatus.php, getRelayState.php, getLiveTemp.php, awaySet.php. SummerSet.php.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
12/05/26	3.23	120526	1.0	050526	Bug Fix - 'request.js', 'db.php' and 'mode.php' – avoid resources issue when clicking on 'Mode', 'Away' and 'Summer Mode' buttons.
18/05/26	3.23	180526	1.0	050526	Bug Fix - 'mode1.php' – Settings/System Configuration/Hot Water Comp not detecting 'Hot Water'zone. Bug Fix – 'controller.py' – disable any running schedules when NOT in Timer Mode.
31/05/26	3.23	310526	1.0	050526	Bug Fix - 'zone.php' – When editing zone, screen not displaying multiple sensors correctly. Bug Fix – 'controller.py' – summer disable schedules not showing correctly. Update – 'db.php' and 'gateway.py' – allow 'anonymous' connection to MQTT broker.
02/06/26	3.23	020626	1.0	020626	New Feature – 'pvvx_ble_mqtt.py', 'check_pvvx_bridge.php' and update to table 'Jobs'– Additional support for 'Xiaomi Mijia Thermometer 2'. Previously support through the use a an ESP32 Bluetooth gateway, this feature enables connection using the Raspberry Pi's Bluetooth adapter, providing a simplified solution.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
20/06/26	3.23	200626	1.0	020626	Update – 'controller.py'– Only display over temperature indicator when a Zone is active, set Zone Mode to 0 when a Zone is inactive, even if over temperature. New Feature – 'getOutsideTemperatureRTI.php'– API to return the min, max, and last 1 hour delta values for the 'Weather Channel' outside temperature. Can also be used with a parameter 'sensorname' to return the values for any other sensor. Bug Fix – 'zone.php'– Ensure that Primary Sensor field remains empty after Zone Type re-selection.
28/06/26	3.23	280626	1.0	020626	Bug Fix – 'controller.py'– Ensure 'Summer Mode' set when zone goes over temperature. 'model.php' and language files – correct display of title for 'Summer Mode' documentation and in 'zone.php'. 'getOutsideTemperatureRTI.php'– fix dor delta values.
30/07/26	3.23	300726	1.0	020626	Bug Fix – 'gateway.py'– fix for MQTT temperature sensors where MQTT payload is numeric.
08/08/26	3.23	080826	1.0	020626	Update – 'ajax_fetch_data.php' – Add script info to 'Settings/System Status/Scripts' GUI for EMS and EBUS boiler control scripts. Updated – 'e7c_control.py' , 'rc10_control.py', 'check_e7c.php' and 'check_ems.php' – Improve restart capability. Bug Fix – 'controller.py' – fix relay synchronization issue for stand alone MySensor relays, only applies when using combined ethernet/relay controller gateway hardware.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
09/08/26	3.23	090826	1.0	020626	Update – 'ajax_fetch_data.php' – Add number of restarts in last 24 hours for Gateway and System Controller scripts to 'Settings/System Status/Scripts' GUI.
10/08/26	3.23	100826	1.0	020626	Update – 'gateway.py', 'controller.py' and 'db_cleanup.py' – replace use of python 'os' library with 'pathlib' and 'subproccess' librarys. Bug Fix – db_cleanup.py' – ensure that only tables present in the database schema are cleared.
11/08/26	3.23	110826	1.0	020626	Update – 'controller.py' – Pause execution if 'db_cleanup.py is running. Add error logging to file '/var/www/logs/controller.log'.
24/08/26	3.23	240826	1.0	020626	Bug Fix – 'controller.py' – One Shot Boost incorrect 'if' indentation. Bug Fix – 'ajax_fetch_data.php' – incorrect use of len(), should be strlen().

Date	Ver	Build	DB Ver	DB Build	ChangeLog
30/08/26	3.23	300826	1.0	300826	Update – 'controller.py', 'gateway.py', 'model.php', 'db.php', 'request.js', '300826.sql' and language files – execute database cleanup and part of 'gateway.py'. rather than as stand alone script 'db_cleanup.py', in order to improve database access synchronisation. Update –'ajax_fetch_data.php' and 'chart_batteryusage.php'– report MQTT/Zigbee sensor battery levels in Settings/System Status/Sensors, display 'No Battery' text for sensors which do not report battery information and improve labelling for battery usage graphs. Bug Fix – 'controller.py' and 'home.php'– 'sc_running' flag not cleared when in Test Mode and clear Test Mode on page timeout return to Home page. Bug Fix – 'gateway.py' – remove calls to 'sys.exit(1)' in functions 'on_disconnect' and 'on_disconnect_2' to allow automatic reconnection.
31/08/26	3.23	310826	1.0	310826	Update – 'relay.php', 'home.php',, 'model.php', 'db.php', 'request.js' and '310826.sql' – improved restore relay states after 'Test Relays', required for stand alone relays, which do not get restored by 'controller.py'. Database update '310826.sql' adds a new column 'previous_state' to the 'relays' table and adds a trigger 'before_relay_state_update', to automatically update the new field.
01/09/26	3.23	010926	1.0	010926	Update – 'home.php', 'db.php' and '010926.sql' – simplification of the restore relay state process, the database update removes the trigger created in the previous update, as no longer required.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
05/09/26	3.23	050926	1.0	050926	Update – 'gateway.py', 'ajax_fetch_data.php', 'check_gw.php', '050926.sql' and language files– tidy up of commit() statements in 'gateway.py', add count of MQTT disconnects in the last 24 hours and update Settings/System Status/Scripts GUI to shown MQTT disconnects, when MQTT is in use. Update – 'mqtt_device.php' – when creating a new MQTT device use any available gaps in the sequence of 'Child IDs' or else append as the next ID.
09/09/26	3.23	090926	1.0	050926	Update – 'controller.py' – reliability improvement, move database connection out of main processing loop, so only connects once on script startup. Bug Fix – 'controller.py' – remove references to 'MQTT_CONNECTED'. Update – 'gateway.py' – change logfile from 'paho-mqtt.log' to 'gateway.log' and add INFO message for 'on_connect' event. Bug Fix – 'ajax_fetch_data.php' – calculation of SUM of all disconnects in the last 24 hours.
10/09/26	3.23	100926	1.0	050926	Bug Fix – 'controller.py' – zone scheduling not detecting disabled zones correctly.
15/09/26	3.23	150926	1.0	050926	New Feature – 'SetRelayState.php' – new API to allow setting of the state of a stand alone relay. Update – 'config_json.py' and 'webhooks.py' – add support for stand alone relays to the 'HomeKit' add on.
16/09/26	3.23	160926	1.0	050926	New Feature – 'mqtt_device.php' and 'gateway.py' – add support for Tuya WiFi type switch for use with 'tinytuya.py' to emulate a local MQTT connection over Ethernet.

Date	Ver	Build	DB Ver	DB Build	ChangeLog
09/10/26	3.23	091026	1.0	091026	Update – 'ajax_fetch_data.php', 'model.php. 'db.php, 'gateway.py' , 'request.js', '091026.sql and language files– add support to 'Sensor Types' model to allow setting of the number of decimal places displayed and to either add, subtract, multiply or divide the displayed value by a definable value. New Feature – tuya_bridge.py' and 'check_tuya_bridge.php' – add script and schedule check support for Tuya WiFi relays. New Feature – model.php' andlanguage files – add new 'Install Software' option for 'tinytuya' Python module. Update – 'gpio_ds18b20.py' – improve error reporting and ensure and MySQL use of 'nodes.node_id' uses a text type parameter. Update – 'rc10_control.py' and 'e7c_control.py' – for sensors where the value does not change, update the 'last_seen' once every 1 minute. Update – 'ajax.php' – set pop-up graph title dependant on the sensor type.